SWEATERCO

GROUP 11 — TRACK 2 / SPRINT 1 REVIEW

The Quick-Response Subcontracting Desk

Who should make this batch? Answered in minutes, with the reasoning on record.

THE NETWORK

QuickStitch
SteadyHands
BudgetWorks
Little Loom
GiantWeave
Nimble Needle
OldMill
FreshStart

OldMill suspended. FreshStart on trial, max 300.

PRIMARY OBJECTIVE **MINIMISE LATENESS**

Chen Zepeng (A0156784N) — Present (in person)
Duong Thanh Nhan (A0296326W) — Present (in person)
Joshua Wesley Gloor (A0296944J) — Present (in person)
Sridhar Niyanthri Rajeswari (A0351151U) — Present (in person)
Sundara Rajan Saadhvika (A0352594W) — Present (in person)
Tang Zheling (A0352177B) — Present (in person)

DSS5105 CAPSTONE / 16 OCTOBER 2026

Today, one chat message is the whole decision

> *@Chen - order 042, 1,500 crewnecks. Send it to QuickStitch, need it back by the 19th.*
>
> HOW ALLOCATION IS DECIDED TODAY / COURSE BRIEF

Everything behind it lives in one dispatcher's head. Nobody can check it, and nothing is written down.

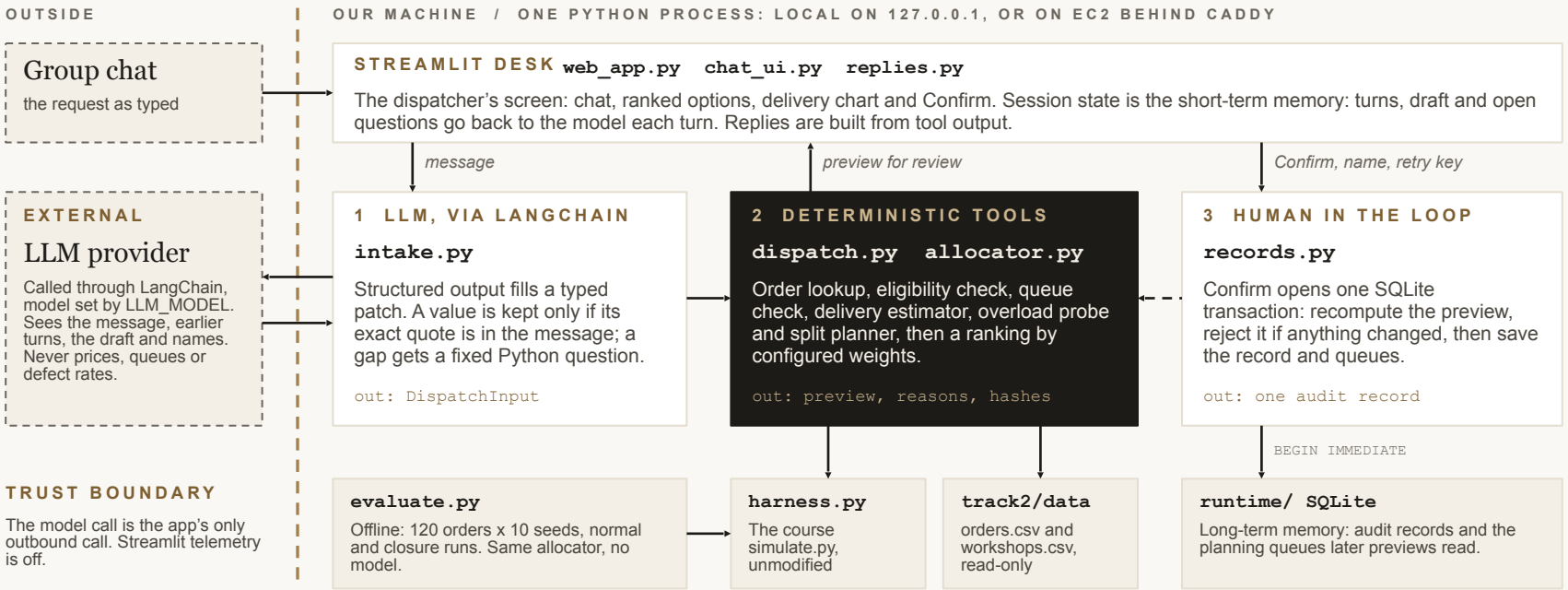
7 of 30 requests in this morning's inbox are plain allocations. The other 23 need a question, a refusal, a constraint or bad news.

THREE QUESTIONS, EVERY TIME

1. **Can they make it?**
   Four of the eight make tops only, so they cannot take a scarf order.
2. **Can they take it now?**
   QuickStitch, "fastest in the region", already has four days of work queued.
3. **Are they allowed to?**
   OldMill is suspended after an audit. FreshStart is on trial, capped at 300 pieces.

Get one wrong and the batch bounces back a day later. In a ten-day cycle, that is the margin gone.

The model reads. Python calculates. A person approves.



Agent pattern today: one well-tooled dispatcher, run as a workflow. The model never picks a tool or does arithmetic.

PRIMARY OBJECTIVE

We declare: minimise lateness

WHY

Quick response sells speed. A defect can be reworked; a missed reorder date loses the sale.

MEASURED BY

The share of batches that arrive late, and the average days late, from the course simulator.

BAR TO CLEAR

Beat random, biggest-capacity and cheapest on both measures, in normal runs and in the closure test.

ALL FOUR COURSE OBJECTIVES: WINS AND LOSSES

OBJECTIVE	OUR BEST SETTING	VERDICT
Minimise lateness	1.7% late, vs 34.2% for random	**Beats all three baselines**
Minimise defects	5.5% defective pieces, 18.4% late	**Biggest-capacity still wins (5.3%)**
Robust to failure	Closure adds 0.25 points late; random adds 4.0	**Steadier than random and biggest. Cheapest stays 92% late either way**
Fair allocation	Not tuned yet: the busiest workshop takes 26.2% of pieces	**Random spreads work more evenly (22.6%)**

Course simulator, mean of 10 runs (seeds 5105-5114). The defects setting was chosen on separate development seeds before this run.

95% fewer late batches than random allocation

1.7%

of batches late, 2 in 120, when queue, processing and transport are checked before choosing

34.2%

41 in 120 for a random eligible workshop

Course simulator, 10 runs. Not a factory result.

1 **Operational efficiency & risk reduction**

No bounced batches: the course's own eligibility rule runs before anything is recommended.

2 **Transparency & auditability**

A full record of the request, every option, the estimates, who set each constraint and who approved.

3 **Proactive exception management**

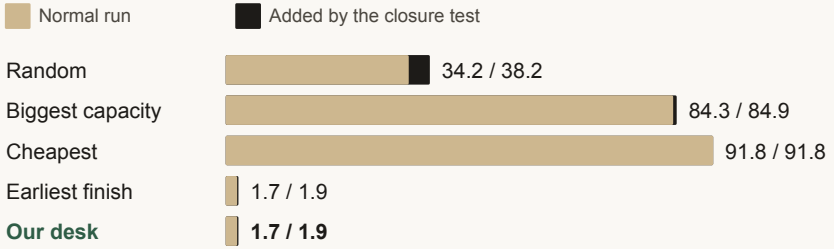
An impossible date gets the earliest achievable date and the options: split, accept late, escalate.

4 **Transparent cost-speed trade-offs**

$115.6k per run, against $86.0k for always-cheapest (late on 92% of batches) and $111.8k for random.

Working allocator, working deployment pipeline

% OF BATCHES LATE, MEAN OF 10 RUNS. LOWER IS BETTER

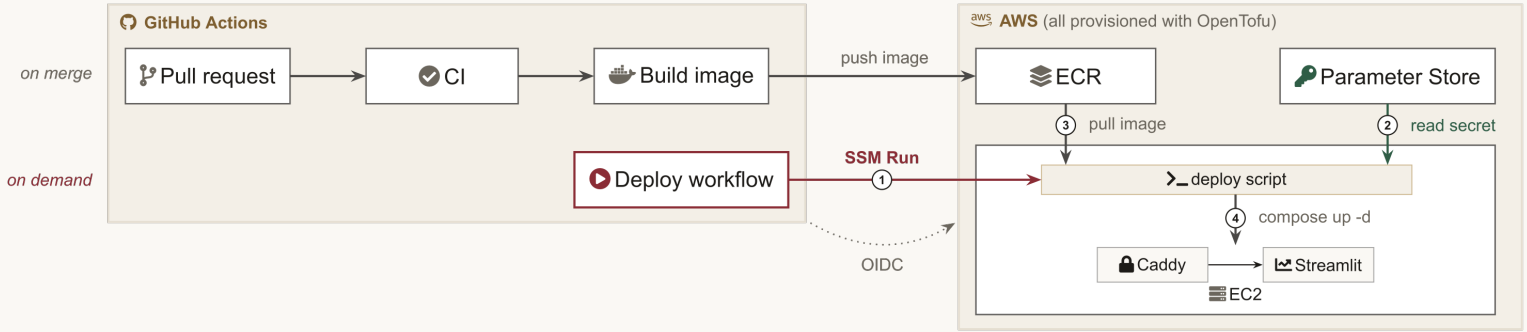


HONEST FINDING

Earliest finish ties us. Lateness is near its floor: two batches of 2,000 accessories cannot be made in their window.

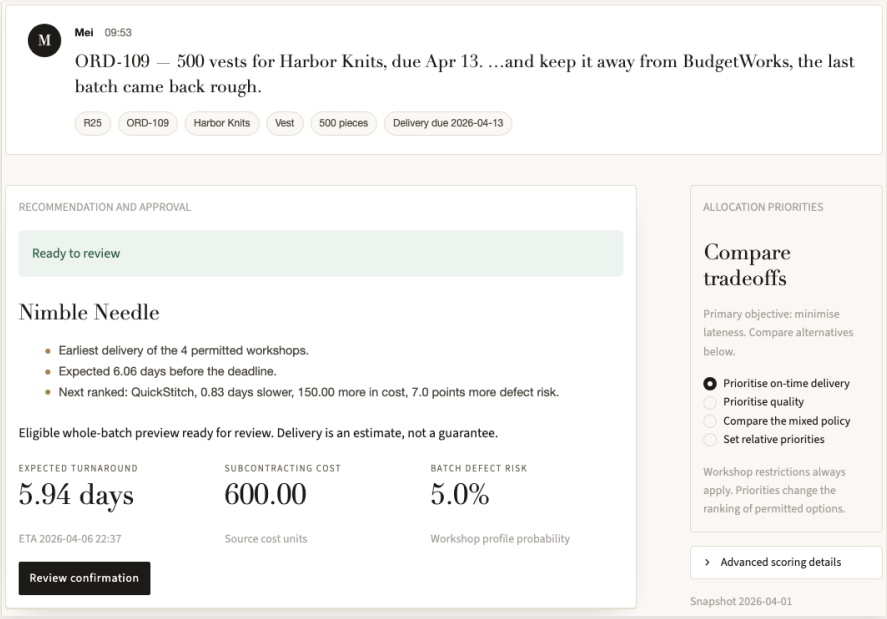
WHAT OUR ALLOCATOR ADDS

The same code, re-weighted: a 10-day buffer cuts cost 7.7%, below random's, and late days 23%, for about one more late batch per ten runs.



Course simulator, seeds 5105-5114; buffer chosen on seeds 6000-6009 first. Sources: docs/V3_EVALUATION.md, docs/MARGIN_STUDY.md, docs/DEPLOYMENT.md (main).

Candidates, estimates and the choice, at a glance



1 **The message stays in view**

Request R25: "keep it away from BudgetWorks". The exclusion is applied and recorded against Mei, who asked.

2 **It says why, in Python's words**

Earliest of the permitted workshops, days to spare, and how the next option compares. No model writes these lines.

3 **Where the days go**

A chart splits each option into queue, processing, transport and rework, so a busy "fast" workshop shows up.

4 **Review before anything is saved**

Confirming rechecks the queues first, then writes one audit record.

ALSO WORKING All 30 inbox requests reach the right outcome from hand-checked inputs / live chat that asks when unsure / stale-safe saved plan / 300 offline tests

What could go wrong, and what we assume

RISK	PROBABILITY / IMPACT	MITIGATION
The model misreads a request	**High** / **High**	Every field must quote the message; Python checks it and asks when unsure. First live run: one needless question blocked a refusal.
A message tries to bend the rules	**Medium** / **High**	Eligibility is a Python check the model cannot switch off. A suspended workshop is refused; exclusions cannot be relaxed mid-chat.
A number no tool produced	**Medium** / **High**	Tools format every number, and an automatic check compares each figure on screen with tool output.
Tuning to the 120 supplied orders	**Medium** / **Medium**	Choose settings on separate development seeds, then run the course seeds once.
The simulator is simpler than a factory	**Certain** / **Medium**	Say so, keep its default run as the yardstick, and test the desk with dispatchers.

KEY ASSUMPTIONS

Each inbox request is judged on the 1 April queues, on its own.

Check: The data sets no order for requests; confirm with the course team. **Risk if wrong: Medium**

A defect means half the batch is redone, as in the simulator.

Check: simulate.py adds 1.5x the work days; confirm with dispatchers. **Risk if wrong: Medium**

Workshop prices are our subcontract cost, not the customer's price.

Check: workshops.csv has cost per piece and no selling prices. **Risk if wrong: Low**

A person confirms every allocation. No workshop is contacted automatically.

Check: Only Confirm writes an allocation (records.py). **Risk if wrong: Low**

From working tools to an AI-native desk

	WORK	SUCCESS CRITERION
1	Evaluation set first	30+ questions checked against the data, 10 held out, 5+ injection and bait items
2	Live chat on all 30 inbox requests	Labels two teammates write independently; decision and explanation scored apart
3	Tool calling for questions	"Who can take 1,500 crewnecks?": the model picks a read-only tool, Python answers
4	Grounded explanations and replies	The model writes with placeholders; Python fills every number, nothing auto-sends
5	LLM judge, checked against people	Agreement with both human labellers reported; people stay the ground truth
6	Freeze the second objective and hybrid	Weights fixed on development seeds, metrics on a common scale, one course run
7	Demo video	Request, clarify, refuse, impossible date, confirm

OUR SUCCESS CRITERION Live chat gets the right behaviour on at least 27 of 30 inbox requests against independent labels, with no number on screen that lacks a tool source, while the desk still beats all three baselines.

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Thank you.

Questions are welcome.

THE MODEL READS / PYTHON CALCULATES / A PERSON APPROVES